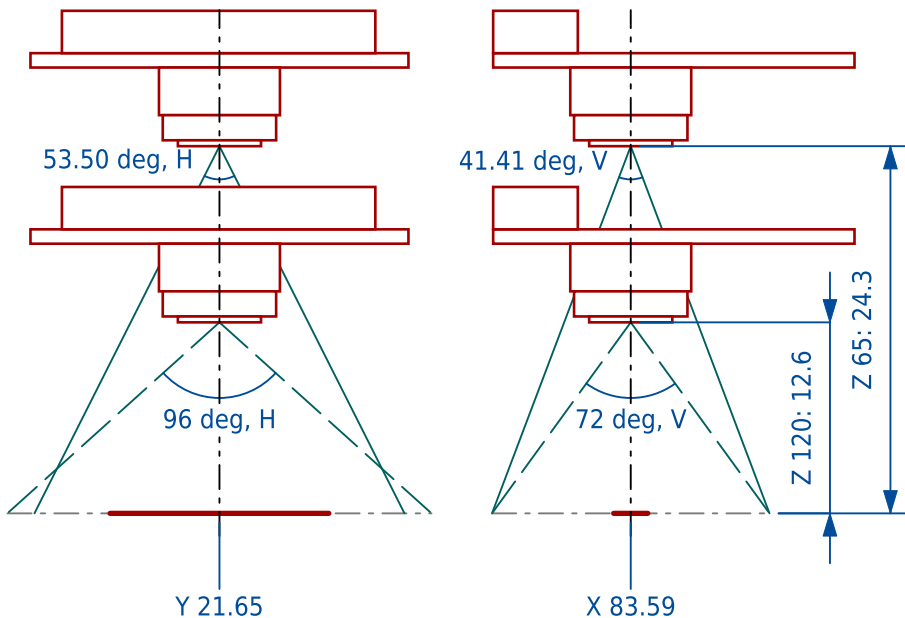
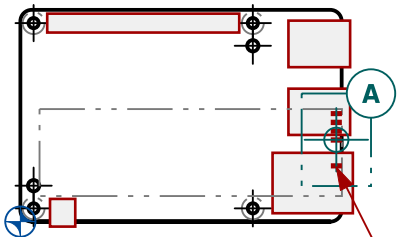


END ELEVATION

FRONT ELEVATION



PLAN, 1:2



NOTES

1. First angle; the end elevation is seen from the right. The camera, a Camera Module v1.3 from its own data, drawn per lens and fitted one at a time, looks straight down, in a mode reading the whole sensor. Z is to its entrance pupil, which nobody locates; ASSUMED behind the lens face by at most the lens's 5.20 mm, so set the FACE at Z and the picture is up to 41.2% larger, never smaller.

LEDs A1-A4 and PWR

NOTES (continued)

2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round, turned whichever way needs the lower camera; LONG says which. One margin absorbs both where the stand ends up -- 5.00 mm of lateral error, or a lean of 10.9 deg at frame A's 65 Z or 17.7 at its 120, measured at the picture's edge -- and what is not known about the lens, below: what one uses the other cannot.
3. PLANE, which Z is measured from: A from the ACORN CARD's top face, not the Pi's -- nobody publishes how far the card stands above the Pi, so measure the stack. Set from the Pi's face the camera sits too low, and the picture at the card loses the ends of it. A stand set h mm too low covers only $(Z - h) / Z$ of what is drawn, so the error always loses the edges.
4. FOCUS. Both fixed lenses as sold are declared "Approx 1 m to infinity" (Raspberry Pi) and "Focus Distance 1 m to infinity" (Arducam), and every Z here is nearer: as sold, the subject is OUT OF FOCUS. At frame A's Z a point spreads to 130 px, 1.0 mm, at 65 and 110 px, 0.7 mm, at 120 (DERIVED, thin lens ASSUMED set at 2 m). Focused closer, Z is the higher of the close limit and $Z + f$ (DERIVED, thin lens); the autofocus module used here publishes NO close limit.

NOTES (continued)

5. The 120 deg lens is a fisheye, 120 deg on the diagonal: 96 x 72 is Arducam's split of that diagonal, equidistant, not a measurement. Its barrel distortion bows the picture's edges outwards on the board, so the frame's corners are inside it; the margin absorbs the narrower lenses the evidence allows, 120 diagonal, equisolid, 94.3 x 69.8, leaving 6.4 mm; Commonlands CIL282, scaled to 120, 94.4 x 70.8, leaving 6.4 mm; YXF4Y001A1, 1.79 mm, 92.4 x 73.9, leaving 5.9 mm. See RPICAM-LENS.
6. The "65 deg" name is the stock lens's DIAGONAL, which no vendor prints: $2 \times \text{atan}(\sqrt{3.6736^2 + 2.7384^2}) / 2 / 3.60 = 64.94$ deg on OmniVision's image area, 64.42 on the pixel array. Every Z here is from the angle along each axis. 65 deg across frame A's long side would give Z 19.2 and lose 2.6 mm off each end.
7. Quotes are transliterated to ASCII: x, um and infinity for the signs the cached pages print.
8. Frame A, A1-A4 and PWR: The card's LEDs, all at its far end: A1 to A4 in a column on the +Y side of the retention screw, PWR on the -Y side. Measured off SQRL's own photograph to +/-1.0 mm, which the margin holds; SQRL published no drawing.

FRAMES, AND Z TO FRAME THEM, mm

	FRAME	RECTANGLE	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	A1-A4 and PWR	18.35 x 24.47	Y	83.59	21.65	24.3	12.6

Z IN FOCUS, mm: THE LOWEST THAT FRAMES AND FOCUSES

	MODULE	CLOSE LIMIT	Z A
v1.3, lens unscrewed by hand	RPi Camera v1.3	60 mm, forum user	60.0
lens focused by hand	Arducam B0031	25.4 mm, DECL	27.9
motorised, the one used here	OV5647 AF-65	not published	--

LENSES, deg, AS USED: SEE RPICAM-LENS

	MODULE	H	V	DIAG	f mm	F	FOCUS FROM
65	RPi Camera v1.3	53.50	41.41	64.4 DERI	3.60 DECL	F2.9 DECL	Fixed, 1 m
AF	OV5647 AF-65	53.50	41.41	64.4 DERI	3.60 ASSU	F2.9 ASSU	Motor, not publ.
120	Arducam B006604	96	72	120.0 DECL	2.17 DERI	F2.4 ASSU	Fixed, 1 m

LEGEND

- Subject outline
- LED, display or connector
- Adjacent part or connector body, and the Acorn CLE-215+
- Frame A's target, on the plane Z is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- The 120 deg lens's field of view, 96 x 72, to its picture's edge
- Frame A, the rectangle the picture must cover
- Dimension, extension and leader

NOTES (continued)

9. The camera is over the LEDs, not the card: the heatsink's blower stands just short of them on the connector side, and from here no ray to an LED passes over it. Nobody publishes the blower's height, and the camera board overhangs it in plan: the board's underside is at $Z + 5.20$, 29.5 at 65, 17.8 at 120 above the card. If the blower stands higher, raise the camera; the picture only grows. The card's seated position is accessories/parts.py's, which ACC-HAT-M2POE is drawn from too.

SOURCES

1. Board geometry: raspberry_pi/boards.py - Pi 5 outline and connectors; see RPI-5.
2. Card, LEDs and HAT: accessories/parts.py - The Acorn, where it is seated and its LEDs, measured off SQRL's photograph by accessories/measure_acorn_leds.py; see ACC-HAT-M2POE, which cites the M.2 specification, SQRL and Waveshare.
3. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

CAMERA OVER THE ACORN CLE-215+

Camera Module OV5647, 65 and 120 degree lenses

DRAWN generated		VERSION v0.0-467-gc00bb8e		UNITS mm		SCALE 2:1, PLAN 1:2	
SIZE A3	SHEET 1 OF 1		DRAWING NO RPICAM-OVER-ACORN				REV A
SUBJECT Acorn CLE-215+, Pi 5			GENERAL TOLERANCE Pi 5 +/-0.20, LEDs +/-1.0 DERIVED, Z DERIVED				PROJECTION 